

Inertial Navigation: IMU Modelling, Kinematics, and Preintegration

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1 Introduction

This document covers the basic inertial measurement unit (IMU) sensor model often used in state estimation literature. IMUs are ubiquitous in robotic state estimation, and have been incorporated into many odometry and SLAM pipelines, including visual-inertial [1], LiDAR-inertial [2], radar-inertial [3], and UWB-inertial [4]. The high-frequency motion information that IMUs provide, along with their relatively cheap cost and their complementary nature with other sensing modalities, make IMUs an attractive option for robust state estimation for a wide range of applications.

Estimation approaches that utilize IMUs can broadly be categorized into two approaches - filtering-based [1], [2], and optimization-based. While filtering-based approaches only linearize the measurement model a single time, optimization-based methods solve a nonlinear least squares problem to update the state iteratively, allowing for the nonlinear measurement models to be relinearized at the cost of higher computational load. Optimization-based approaches have become increasingly popular, both due to the potential higher accuracy compared to filter-based approaches, and the availability of efficient solvers, such as Ceres, GTSAM and g2o.

In a filtering framework, IMU measurements are used to predict the navigation state of the sensor forward in time. However, incorporating IMU measurements into optimization-based approaches presents some challenges, as utilizing IMU measurements directly would result in the addition of a state to the optimization for each measurement. This would quickly become infeasible for real-time estimation with high-rate IMUs. When IMU measurements are naively used form constraints between arbitrary points in time, it requires reintegration of the IMU measurements at each iteration of the optimization, which also becomes computationally infeasible to perform in real-time. To avoid this, the concept of *preintegration* was introduced in [5], to be able to summarize the relative motion constraints between any two arbitrary points in time, in such a way that avoid the need for reintegration of measurements during the optimization.

The main purpose of this document is to summarize how IMU measurements are typically used in both filtering and optimization-based frameworks. Specifically, this document will derive

- The IMU kinematic model in continuous-time, along with different ways of discretizing the IMU kinematics to allow for integrating IMU measurements to propagate forward the navigation state,
- The continuous-time error dynamics to yield process model Jacobians, required for implementing EKF's utilizing an IMU. The process model Jacobians are also required for conducting observability analysis and degenerate motion analysis, key analysis to understand situations where certain parameters may become unobservable. In this document, the navigation state error is defined on the Lie group $SE_2(3)$, which has been shown to better capture the true error dynamics of the IMU navigation state [6].
- IMU preintegration, to showcase how to utilize high-rate IMU measurements efficiently within batch and optimization-based estimation approaches. The notation utilized for preintegration is similar to [6], but this document considers both left and right perturbation def-

initions for the IMU state, as well as propagating the uncertainty of the preintegrated measurements in continuous-time rather than discrete-time.

1.1 Notation and Preliminaries

To showcase the typical IMU sensor model, two frames are introduced - the inertial frame is denoted $\mathcal{F}_a$ and the frame that rotates with the IMU is denoted $\mathcal{F}_b$. An unforced reference particle in $\mathcal{F}_a$ is denoted w and a reference position on the IMU is denoted point z . The direction cosine matrix (DCM) that relates the attitude of $\mathcal{F}_a$ to the attitude of $\mathcal{F}_b$ is denoted $\mathbf{C}_{ab} \in SO(3)$. The DCM relates physical vectors resolved in $\mathcal{F}_a$ to physical vectors resolved in $\mathcal{F}_b$ as $\mathbf{v}_a = \mathbf{C}_{ab}\mathbf{v}_b$. The IMU position and velocity resolved in the inertial frame are denoted $\mathbf{r}_a^{zw}$ and $\mathbf{v}_a^{zw/a}$, respectively.

This document will also make use of Lie group theory, with much of the notation being similar to [7]. A Lie group G is a smooth manifold whose elements, given a group operation $\circ : G \times G \rightarrow G$, satisfy the following group axioms

$$\text{Closure Under } \circ : \quad \mathcal{X} \circ \mathcal{Y} \in G, \quad (1)$$

$$\text{Identity } \mathcal{E} : \quad \mathcal{E} \circ \mathcal{X} = \mathcal{X} \circ \mathcal{E} = \mathcal{X}, \quad (2)$$

$$\text{Inverse} : \quad \mathcal{X} \circ \mathcal{X}^{-1} = \mathcal{X}^{-1} \circ \mathcal{X} = \mathcal{E}, \quad (3)$$

$$\text{Associativity} : \quad (\mathcal{X} \circ \mathcal{Y}) \circ \mathcal{Z} = \mathcal{X} \circ (\mathcal{Y} \circ \mathcal{Z}). \quad (4)$$

For any group G , there exists an associated Lie algebra $\mathfrak{g}$, a vector space identifiable with elements of $\mathbb{R}^m$, where m is the degrees of freedom of the group G . The Lie algebra is related to the Lie group through the exponential and logarithmic maps, denoted $\exp : \mathfrak{g} \rightarrow G$ and $\log : G \rightarrow \mathfrak{g}$. The “vee” and “wedge” operators are denoted $(\cdot)^\vee : \mathfrak{g} \rightarrow \mathbb{R}^m$ and $(\cdot)^\wedge : \mathbb{R}^m \rightarrow \mathfrak{g}$. Similarly to [7], the shortcut $\text{Exp} : \mathbb{R}^m \rightarrow G$ and $\text{Log} : G \rightarrow \mathbb{R}^m$ are defined as

$$\mathcal{X} = \exp(\xi^\wedge) \triangleq \text{Exp}(\xi), \quad (5)$$

$$\xi = \log(\mathcal{X})^\vee \triangleq \text{Log}(\mathcal{X}), \quad (6)$$

where $\xi \in \mathbb{R}^m$.

This document will make use of the $\oplus$ and $\ominus$ operators, allowing for a general way of incrementing elements of Lie groups. There are multiple definitions possible for these operations, and are defined as

$$\mathcal{X} \oplus \delta\xi = \text{Exp}(\delta\xi) \circ \mathcal{X} \quad \text{Lie group left}, \quad (7)$$

$$\mathcal{X} \oplus \delta\xi = \mathcal{X} \circ \text{Exp}(\delta\xi) \quad \text{Lie group right}. \quad (8)$$

Again following [7], the Jacobian of a function $f(\mathcal{X}) : G \rightarrow G$ with respect to $\mathcal{X}$ can be written as

$$\frac{Df(\mathcal{X})}{D\mathcal{X}} \triangleq \lim_{\tau \rightarrow 0} \frac{f(\mathcal{X} \oplus \tau) \ominus f(\mathcal{X})}{\tau}, \quad (9)$$

where either a “left” or “right” definition of the $\oplus$ and $\ominus$ operators may be used. This document will perform all derivations with both “left” and “right” definitions of these operators, allowing for the results to be used in inertial navigation systems that utilize either error definition.

1.2 Lie Group of Extended Poses $SE_2(3)$

In this document, the Lie group $SE_2(3)$ will be utilized to represent *extended poses*, consisting of the attitude, velocity, and position of a rigid body. The attitude, velocity, and position of the body, denoted as $\mathbf{C}_{ab} \in SO(3)$, $\mathbf{v}_a^{zw/a} \in \mathbb{R}^3$, and $\mathbf{r}_a^{zw}$, can be placed into an element of $SE_2(3)$ as

$$\mathbf{T} = \begin{bmatrix} \mathbf{C} & \mathbf{v} & \mathbf{r} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \in SE_2(3) \quad (10)$$

The inverse of an element of $SE_2(3)$ is given by

$$\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{C}^\top & -\mathbf{C}^\top \mathbf{v} & -\mathbf{C}^\top \mathbf{r} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}. \quad (11)$$

The Lie algebra associated with $SE_2(3)$ is denoted $\mathfrak{se}_2(3)$ with

$$\mathfrak{se}_2(3) = \{ \boldsymbol{\xi}^\wedge \in \mathbb{R}^{5 \times 5} | \boldsymbol{\xi} \in \mathbb{R}^9 \}, \quad (12)$$

$$\boldsymbol{\xi}^\wedge = \begin{bmatrix} \boldsymbol{\xi}^\phi \\ \boldsymbol{\xi}^v \\ \boldsymbol{\xi}^r \end{bmatrix} = \begin{bmatrix} \boldsymbol{\xi}^{\phi \times} & \boldsymbol{\xi}^v & \boldsymbol{\xi}^r \\ \mathbf{0} & 0 & 0 \\ \mathbf{0} & 0 & 0 \end{bmatrix}. \quad (13)$$

The exponential map from $\mathfrak{se}_2(3)$ to $SE_2(3)$ is given by

$$\text{Exp}(\boldsymbol{\xi}) = \begin{bmatrix} \text{Exp}(\boldsymbol{\xi}^\phi) & \mathbf{J}_\ell(\boldsymbol{\xi}^\phi) \boldsymbol{\xi}^v & \mathbf{J}_\ell(\boldsymbol{\xi}^\phi) \boldsymbol{\xi}^r \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (14)$$

where $\mathbf{J}_\ell(\boldsymbol{\xi}^\phi)$ is the left Jacobian of $SO(3)$, and $\text{Exp}(\boldsymbol{\xi}^\phi)$ has been overloaded for the exponential map of $SO(3)$. The logarithmic map from $SE_2(3)$ to $\mathfrak{se}_2(3)$ is given by

$$\text{Log}(\mathbf{T}) = \begin{bmatrix} \text{Log}(\mathbf{C}) \\ \mathbf{J}_\ell^{-1}(\text{Log}(\mathbf{C})) \mathbf{v} \\ \mathbf{J}_\ell^{-1}(\text{Log}(\mathbf{C})) \mathbf{r} \end{bmatrix} \in \mathbb{R}^9. \quad (15)$$

The left Jacobian of $SE_2(3)$ is written as

$$\mathbf{J}_\ell(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_\ell(\boldsymbol{\xi}^\phi) & \mathbf{0} & \mathbf{0} \\ \mathbf{Q}(\boldsymbol{\xi}^\phi, \boldsymbol{\xi}^v) & \mathbf{J}_\ell(\boldsymbol{\xi}^\phi) & \mathbf{0} \\ \mathbf{Q}(\boldsymbol{\xi}^\phi, \boldsymbol{\xi}^r) & \mathbf{0} & \mathbf{J}_\ell(\boldsymbol{\xi}^\phi) \end{bmatrix}, \quad (16)$$

and the expression for $\mathbf{Q}(\boldsymbol{\xi}^\phi, \boldsymbol{\xi}^v)$ is given by

$$\begin{aligned} \mathbf{Q}(\boldsymbol{\xi}^\phi, \boldsymbol{\xi}^v) = & \frac{1}{2} \boldsymbol{\nu}^\times + \frac{\phi - \sin \phi}{\phi^3} (\boldsymbol{\phi}^\times \boldsymbol{\nu}^\times + \boldsymbol{\phi}^\times \boldsymbol{\nu}^\times + \boldsymbol{\phi}^\times \boldsymbol{\nu}^\times \boldsymbol{\phi}^\times) \\ & + \frac{\phi + 2 \cos \phi - 2}{2\phi^4} (\boldsymbol{\phi}^\times \boldsymbol{\phi}^\times \boldsymbol{\nu}^\times + \boldsymbol{\nu}^\times \boldsymbol{\phi}^\times \boldsymbol{\phi}^\times - 3\boldsymbol{\phi}^\times \boldsymbol{\nu}^\times \boldsymbol{\phi}^\times) \\ & + \frac{2\phi - 3 \sin \phi + \phi \cos \phi}{2\phi^5} (\boldsymbol{\phi}^\times \boldsymbol{\nu}^\times \boldsymbol{\phi}^\times \boldsymbol{\phi}^\times + \boldsymbol{\phi}^\times \boldsymbol{\phi}^\times \boldsymbol{\nu}^\times \boldsymbol{\phi}^\times). \end{aligned} \quad (17)$$

1.3 Inertial Navigation State Definition

In typical inertial navigation problems, the state for estimation is often the IMU orientation $\mathbf{C}_{ab} \in SO(3)$, the IMU velocity $\mathbf{v}_a^{zw/a}$, IMU position $\mathbf{r}_a^{zw}$, collectively referred to as the *navigation* state, as well as biases of the gyroscope and accelerometer, written $\mathbf{b}_b^g, \mathbf{b}_b^a$ respectively. Stacking the IMU biases as

$$\mathbf{b}_b = \begin{bmatrix} \mathbf{b}_b^g \\ \mathbf{b}_b^a \end{bmatrix} \in \mathbb{R}^6, \quad (18)$$

and denoting the IMU navigation state as $\mathbf{T}_{ab} \in SE_2(3)$, the full IMU state is then defined as

$$\mathcal{X} = (\mathbf{T}_{ab}, \mathbf{b}_b) \in SE_2(3) \times \mathbb{R}^6. \quad (19)$$

This is a composite Lie group, where the group composition operation can be defined as

$$\mathcal{X}_1 \circ \mathcal{X}_2 = (\mathbf{T}_{ab_1} \mathbf{T}_{ab_2}, \mathbf{b}_{b_1} + \mathbf{b}_{b_2}). \quad (20)$$

All other important operations are defined analogously, where each individual operation acts separately on the IMU navigation state and the IMU biases.

This definition of the IMU state allows for the definition of the $\oplus$ operator given by

$$\mathcal{X} \oplus \delta \boldsymbol{\xi} = (\mathbf{T}_{ab} \oplus \delta \boldsymbol{\xi}^{\text{nav}}, \mathbf{b}_b + \delta \boldsymbol{\xi}^b), \quad (21)$$

$$\delta \boldsymbol{\xi} = \begin{bmatrix} \delta \boldsymbol{\xi}^{\text{nav}} \\ \delta \boldsymbol{\xi}^b \end{bmatrix} \in \mathbb{R}^{15}, \quad (22)$$

where $\delta \boldsymbol{\xi}^{\text{nav}} \in \mathbb{R}^9$ is the perturbation to the navigation state and $\delta \boldsymbol{\xi}^b \in \mathbb{R}^6$ is the perturbation to the IMU biases. Both the left and right definitions of the $\oplus$ operator can be used in (21), written as

$$\mathbf{T}_{ab} \oplus \delta \boldsymbol{\xi}^{\text{nav}} = \text{Exp}(\delta \boldsymbol{\xi}^{\text{nav}}) \mathbf{T}_{ab}, \quad \text{left perturbation}, \quad (23)$$

$$\mathbf{T}_{ab} \oplus \delta \boldsymbol{\xi}^{\text{nav}} = \mathbf{T}_{ab} \text{Exp}(\delta \boldsymbol{\xi}^{\text{nav}}), \quad \text{right perturbation}. \quad (24)$$

Additionally, the $\ominus$ operator for the state is defined analogously as

$$\mathcal{X}_1 \ominus \mathcal{X}_2 = \begin{bmatrix} \mathbf{T}_{ab_1} \ominus \mathbf{T}_{ab_2} \\ \mathbf{b}_{b_1} - \mathbf{b}_{b_2} \end{bmatrix} \in \mathbb{R}^{15}, \quad (25)$$

where a left or right definition of the $\ominus$ operator can be used for the navigation state.

2 IMU Sensor Modelling and Kinematics

This document focuses on a basic IMU sensor model that adds bias and noise to the true angular velocity and acceleration of the vehicle. While it is possible to model the intrinsics of the IMU, as done in [1], this is beyond the scope of this document. The rate gyroscope measures

$$\mathbf{u}_b^g(t) = \boldsymbol{\omega}_b^{ba}(t) + \mathbf{b}_b^g(t) + \mathbf{w}_b^g(t), \quad (26)$$

where $\boldsymbol{\omega}_b^{ba}$ is the true angular velocity of the platform, $\mathbf{b}_b^g$ is the additive gyro bias resolved in the body frame, and $\mathbf{w}_b^g$ is white noise of the form $\mathbf{w}_b^g \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c^g \delta(t - \tau))$, where $\mathbf{Q}_c^g$ is the power spectral density of the noise. The accelerometer measurements are modelled to be of the form

$$\mathbf{u}_b^a(t) = \mathbf{C}_{ab}(t)^\top (\mathbf{a}_a^{zw/a/a}(t) + \mathbf{g}_a) + \mathbf{b}_b^a(t) + \mathbf{w}_b^a(t), \quad (27)$$

where $\mathbf{a}_a^{zw/a/a}(t)$ is the true acceleration of the IMU resolved in the inertial frame $\mathcal{F}_a$, and $\mathbf{g}_a$ is the gravity vector resolved in $\mathcal{F}_a$, written as $\mathbf{g}_a = [0 \ 0 \ -g]^\top$, where $g = 9.81\text{m/s}^2$.

Additionally, the gyro bias and accelerometer bias are modelled as first order random walk processes, as

$$\dot{\mathbf{b}}_b^g = \mathbf{w}_b^{b_g}, \quad (28)$$

$$\dot{\mathbf{b}}_b^a = \mathbf{w}_b^{b_a}, \quad (29)$$

where $\mathbf{w}_b^{b_g} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c^{b_g} \delta(t - \tau))$, and $\mathbf{w}_b^{b_a} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c^{b_a} \delta(t - \tau))$.

2.1 Continuous-Time IMU Kinematics

The IMU navigation state consists of the IMU orientation, velocity, and position, written as $\mathcal{X}^{\text{nav}} = (\mathbf{C}_{ab}, \mathbf{v}_a^{zw/a}, \mathbf{r}_a^{zw})$. The time evolution of the IMU navigation state can be written in terms of the true angular velocity and acceleration of the platform, as

$$\dot{\mathbf{C}}_{ab}(t) = \mathbf{C}_{ab}(t) \boldsymbol{\omega}_b^{ba}(t)^\times, \quad (30)$$

$$\dot{\mathbf{v}}_a^{zw/a}(t) = \mathbf{C}_{ab}(t) \mathbf{a}_b^{zw/a/a}(t) + \mathbf{g}_a, \quad (31)$$

$$\dot{\mathbf{r}}_a^{zw}(t) = \mathbf{v}_a^{zw/a}(t). \quad (32)$$

where $(\cdot)^\times$ is the skew-symmetric operator, a mapping from $\mathbb{R}^3$ to $\mathfrak{so}(3)$ such that $\mathbf{a}^\times \mathbf{b} = -\mathbf{b}^\times \mathbf{a}$, $\forall \mathbf{a}, \mathbf{b} \in \mathbb{R}^3$. Subbing in the IMU measurements rather than the true angular velocity and body frame acceleration leads to the continuous-time IMU kinematics given by

$$\dot{\mathbf{C}}_{ab} = \mathbf{C}_{ab} (\mathbf{u}_b^g - \mathbf{b}_b^g - \mathbf{w}_b^g)^\times, \quad (33)$$

$$\dot{\mathbf{v}}_a^{zw/a} = \mathbf{C}_{ab} (\mathbf{u}_b^a - \mathbf{b}_b^a - \mathbf{w}_b^a) + \mathbf{g}_a, \quad (34)$$

$$\dot{\mathbf{r}}_a^{zw} = \mathbf{v}_a^{zw/a}, \quad (35)$$

$$\dot{\mathbf{b}}_b^g = \mathbf{w}_b^{b_g}, \quad \dot{\mathbf{b}}_b^a = \mathbf{w}_b^{b_a}. \quad (36)$$

These are the most common form of the continuous-time IMU kinematic equations on flat Earth, used in many navigation problems involving an IMU. This section will follow [6] to showcase how to write these continuous-time motion equations in a more compact form utilizing the Lie group $SE_2(3)$. Denoting the IMU navigation state as $\mathbf{T}(t) \in SE_2(3)$, written as

$$\mathbf{T}(t) = \begin{bmatrix} \mathbf{C}_{ab}(t) & \mathbf{v}_a^{zw/a}(t) & \mathbf{r}_a^{zw} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \in SE_2(3), \quad (37)$$

the solution to the model (33) at any time t , given the initial condition $\mathbf{T}_0 \in SE_2(3)$ is given by

$$\mathbf{T}(t) = \mathbf{\Gamma}_t \mathbf{\Phi}_t(\mathbf{T}_0) \mathbf{\Upsilon}_t, \quad (38)$$

where $\mathbf{\Phi}_t(\cdot)$ and $\mathbf{\Gamma}_t$ depend only on t and are written as

$$\mathbf{\Phi}_t(\mathbf{T}) \triangleq \begin{bmatrix} \mathbf{C} & \mathbf{v} & \mathbf{r} + t\mathbf{v} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad \mathbf{\Gamma}_t \triangleq \begin{bmatrix} \mathbf{1} & t\mathbf{g}_a & \mathbf{g}_a t^2/2 \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (39)$$

where $\mathbf{C} \triangleq \mathbf{C}_{ab}$, $\mathbf{v} \triangleq \mathbf{v}_a^{zw/a}$ and $\mathbf{r} \triangleq \mathbf{r}_a^{zw}$, for brevity. The quantity $\mathbf{\Upsilon}_t$ is given by

$$\mathbf{\Upsilon}_t \triangleq \begin{bmatrix} \Delta\mathbf{C}_t & \Delta\mathbf{v}_t & \Delta\mathbf{r}_t \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \in SE_2(3), \quad (40)$$

where $\Delta\mathbf{C}_t$, $\Delta\mathbf{v}_t$, and $\Delta\mathbf{r}_t$ are *relative motion increments* defined as

$$\Delta\mathbf{C}_t = \exp\left(\int_0^t \boldsymbol{\omega}_b^{ba} ds\right) = \exp\left(\int_0^t (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g) ds\right), \quad (41)$$

$$\Delta\mathbf{v}_t = \int_0^t \Delta\mathbf{C}_s \mathbf{a}_b ds = \int_0^t \Delta\mathbf{C}_s (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) ds, \quad (42)$$

$$\Delta\mathbf{r}_t = \int_0^t \Delta\mathbf{v}_s ds = \int_0^t \int_0^s \Delta\mathbf{C}_u (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) du ds. \quad (43)$$

These quantities are referred to as *preintegrated measurements* in [5], and are quantities that depend solely on the inertial measurements and do not depend on the initial state $\mathbf{T}_0$. To confirm that (38) is the solution to the the ODEs in (33), consider expanding (38) as

$$\mathbf{T}(t) = \begin{bmatrix} \mathbf{C}(t) & \mathbf{v}(t) & \mathbf{r}(t) \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} = \mathbf{\Gamma}_t \mathbf{\Phi}_t(\mathbf{T}_0) \mathbf{\Upsilon}_t \quad (44)$$

$$= \begin{bmatrix} \mathbf{1} & t\mathbf{g}_a & \mathbf{g}_a \frac{t^2}{2} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{C}_0 & \mathbf{v}_0 & \mathbf{r}_0 + \mathbf{v}_0 t \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\mathbf{C}_t & \Delta\mathbf{v}_t & \Delta\mathbf{r}_t \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (45)$$

$$\mathbf{C}(t) = \mathbf{C}_0 \Delta\mathbf{C}_t(t) \quad (46)$$

$$\mathbf{v}(t) = \mathbf{C}_0 \Delta\mathbf{v}_t(t) + \mathbf{v}_0 + t\mathbf{g}_a, \quad (47)$$

$$\mathbf{r}(t) = \mathbf{C}_0 \Delta\mathbf{r}_t(t) + \mathbf{r}_0 + t\mathbf{v}_0 + \frac{1}{2}\mathbf{g}_a t^2 \quad (48)$$

The time derivative of $\mathbf{T}(t)$ is a function of the time derivatives of the relative motion increments, given by

$$\Delta \dot{\mathbf{C}}_t = \frac{d}{dt} \left(\exp \left(\int_0^t (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times ds \right) \right) \quad (49)$$

$$= \Delta \mathbf{C}_t (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times, \quad (50)$$

$$\Delta \dot{\mathbf{v}}_t = \Delta \mathbf{C}_t (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a), \quad (51)$$

$$\Delta \dot{\mathbf{r}}_t = \Delta \mathbf{v}_t, \quad (52)$$

Taking the time derivative of each component of $\mathbf{T}(t)$ yields

$$\dot{\mathbf{C}}(t) = \mathbf{C}_0 \Delta \dot{\mathbf{C}}_t(t), \quad (53)$$

$$= \mathbf{C}_0 \Delta \mathbf{C}_t (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times, \quad (54)$$

$$= \mathbf{C}(t) (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times \quad (55)$$

$$\dot{\mathbf{v}}(t) = \mathbf{C}_0 \Delta \dot{\mathbf{v}}_t, \quad (56)$$

$$= \mathbf{C}_0 \Delta \mathbf{C}_t(t) (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g}_a \quad (57)$$

$$= \mathbf{C}(t) (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g}_a, \quad (58)$$

$$\dot{\mathbf{r}}(t) = \mathbf{C}_0 \Delta \dot{\mathbf{r}}_t + \mathbf{v}_0 + \frac{1}{2} \mathbf{g}_a t \quad (59)$$

$$= \mathbf{C}_0 \Delta \mathbf{v}(t) + \mathbf{v}_0 + \frac{t}{2} \mathbf{g}_a \quad (60)$$

$$= \mathbf{v}(t). \quad (61)$$

Hence, this confirms that when differentiated, $\mathbf{T}(t) = \Gamma_t \Phi_t(\mathbf{T}_0) \Upsilon_t$ yields the same continuous-time model as in (33).

Next, discrete times $t = t_k$ and $t = t_{k+1}$ will be considered, such that $\Delta t = t_{k+1} - t_k$. Denoting $\Gamma_k \triangleq \Gamma_{\Delta t}$, $\Phi \triangleq \Phi_{\Delta t}$, and $\Upsilon_k \triangleq \Upsilon_{\Delta t}$, extended poses and biases at times k and $k + 1$ are related via

$$\mathbf{T}_{k+1} = \Gamma_k \Phi(\mathbf{T}_k) \Upsilon_k, \quad (62)$$

where Γ_k , $\mathbf{T}_i$, and Υ_k are all elements of $SE_2(3)$. With this definition $\Phi : SE_2(3) \rightarrow SE_2(3)$, note that $\Phi(\mathbf{T}_i \mathbf{T}_j) = \Phi(\mathbf{T}_i) \Phi(\mathbf{T}_j)$, and $\Phi(\text{Exp}(\xi)) = \text{Exp}(\mathbf{F}_{\Delta t} \xi)$, where

$$\mathbf{F}_{\Delta t} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \Delta t \mathbf{1} & \mathbf{1} \end{bmatrix} \in \mathbb{R}^{9 \times 9}. \quad (63)$$

These two important properties can be combined to yield

$$\Phi(\mathbf{T} \text{Exp}(\xi)) = \Phi(\mathbf{T}) \text{Exp}(\mathbf{F}_{\Delta t} \xi), \quad (64)$$

$$\Phi(\text{Exp}(\xi) \mathbf{T}) = \text{Exp}(\mathbf{F}_{\Delta t} \xi) \Phi(\mathbf{T}), \quad (65)$$

which will be particularly useful when deriving Jacobians of the IMU preintegration residuals.

The equation (62) describes how the IMU navigation state evolves over time. To describe the evolution of the biases, consider stacking the biases as $\mathbf{b}_k = [\mathbf{b}_k^g \quad \mathbf{b}_k^a]^\top$, and stacking the random walk noises as $\mathbf{w}^b = [\mathbf{w}_g^{b_g} \quad \mathbf{w}_a^{b_a}]^\top$. Then, the evolution of the biases can be written as

$$\mathbf{b}_{k+1} = \mathbf{b}_k + \int_{t_k}^{t_{k+1}} \mathbf{w}^b ds. \quad (66)$$

Note that together, (62) and (66), describe the exact evolution of the original continuous-time model in (33). However, it does involve integrals that must be computed numerically when trying to predict the IMU state forward in time as measurements arrive. The following sections will derive ways to propagate both the mean and covariance of an extended pose, making different assumptions on which quantities are constant over the integration interval.

2.2 Discrete-Time Mean Propagation

This section will derive several different ways of propagating forward the mean state through the model (62). To evaluate the integrals in (62), common assumptions include

- Piecewise constant global inertial acceleration, as assumed in [5]. This is the most common assumption used when utilizing IMU measurements, but it may often be violated in practice.
- Piecewise constant IMU measurements, which assumes that $\mathbf{u}^g$ and $\mathbf{u}^a$ are constant over the integration interval. This integration scheme leads to an exact closed-form expression for the IMU kinematics and can be found in [6], [8].

2.2.1 Piecewise Constant Global Acceleration

Under the assumption that the global acceleration and angular velocity are constant over the integration interval $t \in [t_k, t_{k+1}]$, the integration components $\Delta \mathbf{C}_{kk+1}$, $\Delta \mathbf{v}_{kk+1}$, and $\Delta \mathbf{r}_{k,k+1}$ can be written as

$$\Delta \mathbf{C}_{kk+1} = \text{Exp} \left(\int_k^{k+1} \boldsymbol{\omega}_b^{ba} ds \right) \quad (67)$$

$$= \text{Exp} (\Delta t \boldsymbol{\omega}_k), \quad (68)$$

$$\Delta \mathbf{v}_{kk+1} = \int_k^{k+1} \Delta \mathbf{C}_s \mathbf{a}_b ds, \quad (69)$$

$$= \int_k^{k+1} \Delta \mathbf{C}_0 \mathbf{a}_b(s) ds, \quad (70)$$

$$= \mathbf{a}_b \Delta t, \quad (71)$$

$$\Delta \mathbf{r}_{kk+1} = \int_k^{k+1} \Delta \mathbf{v}_s ds, \quad (72)$$

$$= \frac{1}{2} \Delta t^2 \mathbf{a}_b, \quad (73)$$

where $\Delta t = t_{k+1} - t_k$ and $\mathbf{a}_b \triangleq \mathbf{a}_b^{zw/a/a}$ is the inertial acceleration of the platform resolved in the body frame. With this definition of the integration components, the RMI Υ_k can be written as

$$\Upsilon_k = \begin{bmatrix} \text{Exp}(\boldsymbol{\omega}_k \Delta t) & \mathbf{a}_{b_k} \Delta t & \frac{1}{2} \Delta t^2 \mathbf{a}_{b_k} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (74)$$

and the discrete-time IMU kinematics can be written as

$$\mathbf{T}_{k+1} = \mathbf{\Gamma}_k \Phi(\mathbf{T}_k) \Upsilon_k \quad (75)$$

$$= \begin{bmatrix} \mathbf{1} & \Delta t \mathbf{g}_a & \frac{1}{2} \Delta t^2 \mathbf{g}_a \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{C}_k & \mathbf{v}_k & \mathbf{r}_k + \mathbf{v}_k \Delta t \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \begin{bmatrix} \text{Exp}(\boldsymbol{\omega}_k \Delta t) & \mathbf{a}_{b_k} \Delta t & \frac{1}{2} \Delta t^2 \mathbf{a}_{b_k} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (76)$$

$$= \begin{bmatrix} \mathbf{C}_{k+1} & \mathbf{v}_{k+1} & \mathbf{r}_{k+1} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (77)$$

where

$$\mathbf{C}_{k+1} = \mathbf{C}_k \text{Exp}(\boldsymbol{\omega}_k \Delta t), \quad (78)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + (\mathbf{C}_k \mathbf{a}_{b_k} + \mathbf{g}_a) \Delta t, \quad (79)$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k + \Delta t \mathbf{v}_k + \frac{1}{2} (\mathbf{C}_k \mathbf{a}_{b_k} + \mathbf{g}_a) \Delta t^2, \quad (80)$$

where the sensor models $\boldsymbol{\omega}_k = \mathbf{u}_k^g - \mathbf{b}_k^g - \mathbf{w}_k^g$, and $\mathbf{a}_{b_k} = \mathbf{u}_k^a - \mathbf{b}_k^a - \mathbf{w}_k^a$ can be substituted in to write the discrete-time IMU kinematics as a function of measurements. Finally, discretizing the bias dynamics assuming that the noise is constant over the integration interval leads to

$$\mathbf{b}_{k+1}^g = \mathbf{b}_k^g + \Delta t \mathbf{w}_k^{b_g}, \quad (81)$$

$$\mathbf{b}_{k+1}^a = \mathbf{b}_k^a + \Delta t \mathbf{w}_k^{b_a}. \quad (82)$$

This is the most commonly seen discrete-time IMU kinematic model, and is for example utilized in [5].

2.2.2 Piecewise Constant IMU Measurements

This section will derive that the IMU measurements $\mathbf{u}^g$ and $\mathbf{u}^a$ themselves are constant over the integration interval. The attitude discretization is identical to the previous section, and

$$\mathbf{C}_{k+1} = \mathbf{C}_k \text{Exp}(\boldsymbol{\omega}_k \Delta t). \quad (83)$$

For the velocity dynamics, assuming that $\mathbf{a}_b(t) = \mathbf{u}^a(t) - \mathbf{b}^a(t) - \mathbf{w}^a(t)$ is constant over the interval, the integral becomes

$$\Delta \mathbf{v}_{kk+1} = \int_{t_k}^{t_{k+1}} \Delta \mathbf{C}_s \mathbf{a}_b(s) ds \quad (84)$$

$$= \mathbf{a}_{b_k} \int_{t_k}^{t_{k+1}} \text{Exp}(\boldsymbol{\omega}_k s) ds. \quad (85)$$

Similarly, the position integral becomes

$$\Delta \mathbf{r}_{kk+1} = \int_{t_k}^{t_{k+1}} \int_{t_k}^s \Delta \mathbf{C}_u \mathbf{a}_b(u) du ds \quad (86)$$

$$= \mathbf{a}_{b_k} \int_{t_k}^{t_{k+1}} \int_{t_k}^s \text{Exp}(\boldsymbol{\omega}_k u) du ds. \quad (87)$$

Solving these integrals can be done as shown in [8][p.100] by defining the following auxilarly function:

$$\Psi_m(\phi) \triangleq \left(\sum_{n=0}^{\infty} \frac{1}{(n+m)!} (\phi^{\times n}) \right), \quad (88)$$

which allows these integrals to be expressed using the Taylor series form of the exponential map using

$$\int_{t_k}^{t_{k+1}} \text{Exp}(\boldsymbol{\omega} s) ds = \int \Psi_0(\boldsymbol{\omega}_k s) ds = \Psi_1(\boldsymbol{\omega} \Delta t) \Delta t, \quad (89)$$

$$\int_{t_k}^{t_{k+1}} \int_{t_k}^s \text{Exp}(\boldsymbol{\omega}_k s) du ds = \Psi_2(\boldsymbol{\omega}_k \Delta t) \Delta t^2, \quad (90)$$

where closed-form expressions for $\Psi_0(\cdot)$, $\Psi_1(\cdot)$, and $\Psi_2(\cdot)$ are given by

$$\Psi_0(\phi) = \mathbf{1} + \frac{\sin(\phi)}{\phi} \mathbf{a}^{\times} + \frac{\phi - \cos(\phi)}{\phi^2} \mathbf{a}^{\times 2}, \quad (91)$$

$$\Psi_1(\phi) = \mathbf{1} + \frac{1 - \cos(\phi)}{\phi} \mathbf{a}^{\times} + \frac{\phi - \sin(\phi)}{\phi^2} \mathbf{a}^{\times 2}, \quad (92)$$

$$\Psi_2(\phi) = \frac{1}{2} \mathbf{1} + \frac{\phi - \sin(\phi)}{\phi^3} \mathbf{a}^{\times} + \frac{\phi^2 + 2 \cos(\phi) - 2}{2\phi^4} \mathbf{a}^{\times 2}, \quad (93)$$

where $\phi = \phi \mathbf{a}$, with $\phi = \|\phi\|$ and $\mathbf{a} = \phi/\phi$. Note that $\Psi_0(\phi) = \text{Exp}(\phi)$ is simply the exponential map of $SO(3)$, while $\Psi_1(\phi)$ is the left Jacobian of $SO(3)$. With these definitions of the integration components, Υ_k can be written as

$$\Upsilon_k = \begin{bmatrix} \text{Exp}(\boldsymbol{\omega}_k \Delta t) & \Delta t \mathbf{J}_\ell(\Delta t \boldsymbol{\omega}_k) \mathbf{a}_{b_k} & \Delta t^2 \Psi_2(\Delta t \boldsymbol{\omega}_k) \mathbf{a}_{b_k} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (94)$$

where $\mathbf{J}_\ell(\cdot)$ is the left Jacobian of $SO(3)$. Utilizing this definition of Υ leads to the discrete-time IMU given by

$$\mathbf{C}_{k+1} = \mathbf{C}_k \text{Exp}(\boldsymbol{\omega}_k \Delta t), \quad (95)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{C}_k \Psi_1(\boldsymbol{\omega}_k \Delta t) \mathbf{a}_{b_k} \Delta t + \mathbf{g}_a \Delta t, \quad (96)$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k + \mathbf{v}_k + \mathbf{C}_k \Psi_2(\boldsymbol{\omega}_k \Delta t) \mathbf{a}_{b_k} \Delta t^2 + \frac{1}{2} \mathbf{g}_a \Delta t^2. \quad (97)$$

2.3 Continuous-Time Error Dynamics

The continuous-time error dynamics are often needed for state estimation (as is the case the covariance prediction step of continuous-time Kalman filtering), or for analysis purposes, like when performing an observability analysis. This section will outline the forms of the continuous-time error dynamics for both a left and a right perturbation of the IMU navigation state (i.e., for both (23) and (24)).

The continuous-time process model is written as

$$\dot{\mathcal{X}}(t) = (\dot{\mathbf{T}}(t), \dot{\mathbf{b}}(t)) = f(\mathcal{X}(t), \mathbf{u}(t), \mathbf{w}(t)), \quad (98)$$

where the function $f(\mathcal{X}(t), \mathbf{u}(t), \mathbf{w}(t))$ can be written as

$$(\dot{\mathbf{T}}, \dot{\mathbf{b}}) = (f_1(\mathcal{X}(t), \mathbf{u}(t), \mathbf{t}), f_2(\mathcal{X}(t), \mathbf{u}(t), \mathbf{w}(t))), \quad (99)$$

where $f_1(\mathcal{X}(t), \mathbf{u}(t))$ is given by

$$\dot{\mathbf{T}}(t) = \begin{bmatrix} \dot{\mathbf{C}}(t) & \dot{\mathbf{v}}(t) & \dot{\mathbf{r}}(t) \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (100)$$

with $\dot{\mathbf{C}}(t), \dot{\mathbf{v}}(t), \dot{\mathbf{r}}(t)$ given in (33). The continuous-time error dynamics are of the form

$$\delta \dot{\boldsymbol{\xi}}(t) \approx \mathbf{F}_c(t) \delta \boldsymbol{\xi}(t) + \mathbf{L}_c(t) \delta \mathbf{w}(t), \quad (101)$$

where

$$\mathbf{F}_c \triangleq \frac{Df(\mathcal{X}, \mathbf{u}, \mathbf{w})}{D\mathcal{X}}, \quad (102)$$

$$\mathbf{L}_c \triangleq \frac{Df(\mathcal{X}, \mathbf{u}, \mathbf{w})}{D\mathbf{w}}. \quad (103)$$

where the Lie group definition of a Jacobian as in [7] is used. The exact expressions for the Jacobians $\mathbf{F}_c$ and $\mathbf{L}_c$ depend on whether a left or right definition of the $\oplus$ operator is used. Starting with a right definition of the $\oplus$ operator, these Jacobians are given as

$$\mathbf{F}_c = \begin{bmatrix} -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} & \mathbf{0} \\ -(\mathbf{u}^a - \mathbf{b}^a)^\times & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} \\ \mathbf{0} & \mathbf{1} & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_c = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}. \quad (104)$$

For a left definition of the $\oplus$ operator, the continuous-time process model Jacobians are given by

$$\mathbf{F}_c = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{C}_{ab} & \mathbf{0} \\ \mathbf{g}^\times & \mathbf{0} & \mathbf{0} & -\mathbf{v}_a^{zw/a^\times} \mathbf{C}_{ab} & -\mathbf{C}_{ab} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & -\mathbf{r}_a^{zw^\times} \mathbf{C}_{ab} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_c = \begin{bmatrix} \mathbf{C}_{ab} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{v}_a^{zw/a^\times} \mathbf{C}_{ab} & \mathbf{C}_{ab} & \mathbf{0} & \mathbf{0} \\ \mathbf{r}_a^{zw^\times} \mathbf{C}_{ab} & \mathbf{0} & \mathbf{C}_{ab} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}. \quad (105)$$

Note that the continuous-time matrix $\mathbf{F}_c$ depends on the input and biases for a right perturbation definition, whereas it depends on the navigation state for the left perturbation definition. The full derivations for these Jacobians can be found in [Appendix A](#).

To implement a state estimation algorithm like an EKF, an equivalent discrete-time linear system of the form

$$\delta\boldsymbol{\xi}_k \approx \mathbf{F}_{k-1}\delta\boldsymbol{\xi}_{k-1} + \delta\mathbf{w}_{k-1}, \quad (106)$$

is required, where $\delta\mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{k-1})$. To convert from the continuous-time error dynamics in (101) to the discrete-time error dynamics, one common technique is to assume that the continuous-time error dynamics are piecewise time-invariant, and then to use a technique such as Van Loan's method to convert the continuous-time LTI system to a discrete one. Once a discrete-time system of the form (106) is obtained, the covariance in an EKF can be propagated forward using

$$\check{\mathbf{P}}_k = \mathbf{F}_{k-1}\hat{\mathbf{P}}_{k-1}\mathbf{F}_{k-1}^\top + \mathbf{Q}_{k-1}. \quad (107)$$

Note that when utilizing the continuous-time error dynamics to propagate the covariance in an EKF utilizing (107), either discretization scheme can be used in the discrete-time mean propagation step.

3 IMU Preintegration

The previous section derived the IMU kinematic model and demonstrated how to discretize the IMU kinematics to propagate forward the navigation state one step in time, as well as Jacobians that are required to implement EKF and perform analysis with problems involving an IMU. This section will showcase how to perform *IMU preintegration*, used to link the IMU state at any two arbitrary points in time, denoted $t = t_i$ and $t = t_j$.

Consider the relationship between extended poses $\mathbf{T}_i$ and $\mathbf{T}_j$ at arbitrary points in time, given by

$$\mathbf{T}_j = \mathbf{\Gamma}_{ij} \mathbf{\Phi}_{ij}(\mathbf{T}_i) \mathbf{\Upsilon}_{ij}. \quad (108)$$

Rearranging to isolate $\mathbf{\Upsilon}_{ij}$ yields

$$\mathbf{\Upsilon}_{ij} = (\mathbf{\Gamma}_{ij} \mathbf{\Phi}_{ij}(\mathbf{T}_i))^{-1} \mathbf{T}_j, \quad (109)$$

where recall that $\mathbf{\Upsilon}_{ij}$ *only* depends on the measurements obtained in the interval $t \in [t_i, t_j]$, and the right hand side depends on the extended poses $\mathbf{T}_i$, $\mathbf{T}_j$ and the time interval Δt . Expanding the right hand side yields

$$\mathbf{\Phi}_{ij}(\mathbf{T}_i)^{-1} \mathbf{\Gamma}_{ij}^{-1} \mathbf{T}_j = \begin{bmatrix} \Delta \mathbf{C}_{ij} & \Delta \mathbf{v}_{ij} & \Delta \mathbf{r}_{ij} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (110)$$

$$\Delta \mathbf{C}_{ij} = \mathbf{C}_i^\top \mathbf{C}_j, \quad (111)$$

$$\Delta \mathbf{v}_{ij} = \mathbf{C}_i^\top (\mathbf{v}_j - \mathbf{v}_i - \mathbf{g}_a \Delta t_{ij}), \quad (112)$$

$$\Delta \mathbf{r}_{ij} = \mathbf{C}_i^\top \left(\mathbf{r}_j - \mathbf{r}_i - \mathbf{v}_i \Delta t_{ij} - \frac{1}{2} \mathbf{g}_a \Delta t_{ij}^2 \right). \quad (113)$$

There are two expressions for the quantities $\Delta \mathbf{C}_{ij}$, $\Delta \mathbf{v}_{ij}$ and $\Delta \mathbf{r}_{ij}$ - one expressed in terms of the states at times $t = t_i$ and $t = t_j$, and one expressed in terms of the IMU measurements that occurred in the interval $t \in [t_i, t_j]$. In the following, the quantity $\hat{\mathbf{\Upsilon}}_{ij}$ will be used to denote the RMI computed from measurements, while the quantity $\mathbf{\Upsilon}_{ij}(\mathbf{T}_i, \mathbf{T}_j) = (\mathbf{\Gamma}_{ij} \mathbf{\Phi}_{ij}(\mathbf{T}_i))^{-1} \mathbf{T}_j$ will be used to denote the RMI expressed in terms of the states at times $t = t_i$ and $t = t_j$.

The goal is to compute the RMI $\hat{\mathbf{\Upsilon}}_{ij}$ directly from the IMU measurements as they arrive, while also characterizing the uncertainty on $\hat{\mathbf{\Upsilon}}_{ij}$ due to measurement noise. Once $\hat{\mathbf{\Upsilon}}_{ij}$ and its uncertainty are computed, they can be used directly as a noisy measurement that links the states at times $t = t_i$ and $t = t_j$, with the *measurement model* $\mathbf{\Upsilon}_{ij}(\mathbf{T}_i, \mathbf{T}_j)$ that allows us to predict the noiseless values of $\mathbf{\Upsilon}_{ij}$ given the states at times $t = t_i$ and $t = t_j$. The following section will derive how to compute $\hat{\mathbf{\Upsilon}}_{ij}$ from measurements as they arrive.

Also note that the expressions for (111) - (113) are identical to those found in [5], meaning that the approach outlined in this document leads to an identical mean definition of the RMIs. The key difference lies in how the uncertainty on the RMI is defined - here, as will be shown, the uncertainty on the RMI is defined on $SE_2(3)$.

3.1 Discrete-Time RMI Computation

To showcase how to compute the RMI from measurements, consider propagating an extended pose through (62) between two consecutive timestamps as

$$\mathbf{T}_{i+2} = \mathbf{\Gamma}_{i+1} \mathbf{\Phi}_{\Delta t}(\mathbf{T}_{i+1}) \mathbf{\Upsilon}_{i+1}, \quad (114)$$

$$= \mathbf{\Gamma}_{i+1} \mathbf{\Phi}_{\Delta t}(\mathbf{\Gamma}_i \mathbf{\Phi}_{\Delta t}(\mathbf{T}_i) \mathbf{\Upsilon}_i) \mathbf{\Upsilon}_{i+1}, \quad (115)$$

$$= \underbrace{\mathbf{\Gamma}_{i+1} \mathbf{\Phi}_{\Delta t}(\mathbf{\Gamma}_i)}_{\mathbf{\Phi}_{ii+2}} \underbrace{\mathbf{\Phi}_{\Delta t}(\mathbf{T}_i)}_{\mathbf{\Phi}_{ii+2}(\mathbf{T}_i)} \underbrace{\mathbf{\Phi}_{\Delta t}(\mathbf{\Upsilon}_i) \mathbf{\Upsilon}_{i+1}}_{\hat{\mathbf{\Upsilon}}_{ii+2}}, \quad (116)$$

Utilizing this, the incremental formula for how to compute the RMI as new measurements arrive is given by

$$\hat{\mathbf{\Upsilon}}_{ij} = \prod_{k=i}^{j-1} \mathbf{\Phi}_{k(j-1)}(\hat{\mathbf{\Upsilon}}_k), \quad (117)$$

where $\hat{\mathbf{\Upsilon}}_k$ is computed from IMU measurements at time t_k using either the constant inertial acceleration assumption or the constant IMU measurement assumption, as shown in (74) and (94) respectively.

3.2 Including IMU Biases

When computing the RMI from measurements, the IMU biases also enter the computation in (74) and (94). Consider definition a bias RMI as

$$\Delta \mathbf{b}_{ij}(\mathbf{b}_i, \mathbf{b}_j) = \mathbf{b}_j - \mathbf{b}_i, \quad (118)$$

which may also be written by integrating the noise from $t = t_i$ to $t = t_j$ as

$$\Delta \mathbf{b}_{ij} = \int_{t_i}^{t_j} \mathbf{w}^b d\tau. \quad (119)$$

With these definitions, the RMI including the biases can be written as

$$\Delta \mathcal{X}_{ij} = (\mathbf{\Upsilon}_{ij}, \Delta \mathbf{b}_{ij}) \in SE_2(3) \times \mathbb{R}^6. \quad (120)$$

In the subsequent sections, the RMI computed as a function of the states will be written as $\Delta \mathcal{X}_{ij}$, whereas the noisy RMI computed from measurements will be denoted as $\Delta \hat{\mathcal{X}}_{ij}$.

3.3 RMI Uncertainty Definition

As the RMI $\Delta \hat{\mathcal{X}}_{ij}$ is computed from noisy IMU measurements, it is itself an uncertain quantity, and characterizing the uncertainty on the RMI is critical to utilizing it as a measurement in state estimation problems. The noise on the RMI is defined as

$$\Delta \mathcal{X}_{ij} = \hat{\mathcal{X}}_{ij} \oplus \boldsymbol{\eta}_{ij}, \quad (121)$$

where $\boldsymbol{\eta}_{ij} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{ij})$, and the $\oplus$ operator may be defined in a left or right manner for the quantity $\boldsymbol{\Upsilon}_{ij} \in SE_2(3)$. Regardless of the choice of perturbation direction, characterizing the covariance $\boldsymbol{\Sigma}_{ij}$ is critical. The following sections will derive how the covariance $\boldsymbol{\Sigma}_{ij}$ may be computed recursively, as new IMU measurements arrive. While the original derivations in [5] and [6] derive the covariance propagation in discrete-time, this document will consider the continuous-time evolution of the RMI in order to propagate forward the covariance.

3.4 Continuous-Time RMI Dynamics

The continuous-time RMI kinematics can be found by differentiating the definitions of the individual RMIs with respect to time, as done in (49) - (52). These quantities are given by

$$\Delta \dot{\mathbf{C}}_{it} = \Delta \mathbf{C}_{it} (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^a)^\times, \quad (122)$$

$$\Delta \dot{\mathbf{v}}_{it} = \Delta \mathbf{C}_{it} (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a), \quad (123)$$

$$\Delta \dot{\mathbf{r}}_{it} = \Delta \mathbf{v}_{it}, \quad (124)$$

$$\Delta \dot{\mathbf{b}}_{it} = \mathbf{w}_b^b, \quad (125)$$

where note that rather than choosing an explicit end time $t = t_j$, the RMIs are computed from an initial time $t = t_i$ to a varying time t . Similarly to the continuous-time IMU kinematics linearization shown in Section 2.3, the continuous-time RMI kinematics can now be linearized. The linearization of the RMI kinematics leads to an equation of the form

$$\delta \dot{\boldsymbol{\xi}}^{\Delta \mathcal{X}} \approx \mathbf{F}_c^{\Delta \mathcal{X}} \Delta \boldsymbol{\xi}^{\Delta \mathcal{X}} + \mathbf{L}_c^{\Delta \mathcal{X}} \Delta \mathbf{w}, \quad (126)$$

where $\delta \boldsymbol{\xi}^{\Delta \mathcal{X}}$ represents the RMI error. The RMI error can be defined analogously to the state error, by noting that the RMI $\Delta \mathcal{X}_{ij} \in SE_2(3) \times \mathbb{R}^6$, allowing for the definitions of the $\oplus$ and $\ominus$ operators as in (21) and (25). As previously, either a left or right definition of these operators can be used, leading to different Jacobians in (126). Starting with a left definition of the $\oplus$ operator, the Jacobians in (126) are given by

$$\mathbf{F}_c^{\Delta \mathcal{X}} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\Delta \bar{\mathbf{C}}_{it} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\Delta \bar{\mathbf{v}}_{it}^\times \Delta \bar{\mathbf{C}}_{it} & -\Delta \bar{\mathbf{C}}_{it} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & -\Delta \mathbf{r}_{it}^\times \Delta \mathbf{C}_{it} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_c^{\Delta \mathcal{X}} = \begin{bmatrix} \Delta \bar{\mathbf{C}}_{it} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \Delta \bar{\mathbf{v}}_{it}^\times \Delta \bar{\mathbf{C}}_{it} & \Delta \bar{\mathbf{C}}_{it} & \mathbf{0} & \mathbf{0} \\ \Delta \mathbf{r}_{it}^\times \Delta \mathbf{C}_{it} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}. \quad (127)$$

For a right definition of the $\oplus$ operator, the continuous-time process model Jacobians are given by

$$\mathbf{F}_c^{\Delta \mathcal{X}} = \begin{bmatrix} -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} & \mathbf{0} \\ -(\mathbf{u}^a - \mathbf{b}^a) & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} \\ \mathbf{0} & \mathbf{1} & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_c^{\Delta \mathcal{X}} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}. \quad (128)$$

The derivation of these Jacobians is very similar to the derivation of the continuous-time IMU kinematics linearization, shown in Appendix A.

As with the IMU kinematic linearization, the continuous-time RMI error dynamics can be discretized using any method of choice, to yield a discrete-time linear dynamics given by

$$\delta \boldsymbol{\xi}_{k-1}^{\Delta \mathcal{X}} \approx \mathbf{F}_{k-1}^{\Delta \mathcal{X}} \delta \boldsymbol{\xi}_{k-1}^{\Delta \mathcal{X}} + \mathbf{L}_{k-1}^{\Delta \mathcal{X}} \delta \mathbf{w}_{k-1}. \quad (129)$$

3.5 IMU Preintegration Summary

To summarize, IMU preintegration involves recursively propagating the mean RMI $\Delta \hat{\mathcal{X}}_{ij}$ and the covariance $\boldsymbol{\Sigma}_{ij}$ forward as new IMU measurements arrive. Starting from an initial mean $\Delta \hat{\mathcal{X}}_{ii} = (\mathbf{1}, \mathbf{0})$ and covariance $\boldsymbol{\Sigma}_{ii} = \mathbf{0}$, given an IMU measurement at time $t_{k-1} \in [t_i, t_j]$, the following steps are taken.

- Propagate forward the mean using the discrete-time equation

$$\hat{\boldsymbol{\Upsilon}}_{ik} = \Phi_{\Delta t} \left(\hat{\boldsymbol{\Upsilon}}_{ik-1} \right) \hat{\boldsymbol{\Upsilon}}_k, \quad (130)$$

where $\hat{\boldsymbol{\Upsilon}}_k$ is computed from the IMU measurements utilizing either the piecewise constant acceleration assumption in (74) or the piecewise constant IMU measurement assumption in (94).

- Propagate forward the covariance using

$$\boldsymbol{\Sigma}_{ik} = \mathbf{F}_{k-1}^{\Delta \mathcal{X}} \boldsymbol{\Sigma}_{ik-1} \mathbf{F}_{k-1}^{\Delta \mathcal{X}^\top} + \mathbf{Q}_{d_{k-1}}, \quad (131)$$

where $\mathbf{F}_{k-1}^{\Delta \mathcal{X}}$ and $\mathbf{Q}_{k-1}^{\Delta \mathcal{X}}$ are the result of discretizing the continuous-time linear dynamics in (126), with timestep $\delta t = t_k - t_{k-1}$.

4 Batch Estimation Preintegration Residual

This section will explore how to use the preintegrated RMI, $\Delta\hat{\mathcal{X}}_{ij}$ and covariance Σ_{ij} , computed from measurements as described in the previous section, within the context of batch estimation. Defining the RMI computed from states as

$$\Delta\mathcal{X}_{ij}(\mathcal{X}_i, \mathcal{X}_j) = (\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j), \Delta\mathbf{b}_{ij}), \quad (132)$$

$$\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) = (\Gamma_{ij}\Phi_{ij}(\mathbf{T}_i))^{-1}\mathbf{T}_j, \quad (133)$$

$$\Delta\mathbf{b}_{ij} = \mathbf{b}_j - \mathbf{b}_i, \quad (134)$$

the preintegrated residual can be defined as

$$\mathbf{e}_{ij}(\mathcal{X}_i, \mathcal{X}_j) = \Delta\mathcal{X}_{ij}(\mathcal{X}_i, \mathcal{X}_j) \ominus \Delta\hat{\mathcal{X}}_{ij} \quad (135)$$

$$= \begin{bmatrix} \Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) \ominus \hat{\Upsilon}_{ij} \\ \mathbf{b}_j - \mathbf{b}_i \end{bmatrix}, \quad (136)$$

where the $\ominus$ operator can be defined in either a left or right fashion for the RMI, which is an element of the group $SE_2(3) \times \mathbb{R}^6$. For example, with a left definition of the $\ominus$ operator, the residual is expressed as

$$\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) \ominus \hat{\Upsilon}_{ij} = \text{Log} \left(\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) \hat{\Upsilon}_{ij}^{-1} \right). \quad (137)$$

The covariance on this error term is simply the covariance of the preintegrated RMI computed from measurements, given by Σ_{ij} .

Recall that the recursive computation of $\Delta\hat{\mathcal{X}}_{ij}$ depends *both* on the IMU measurements and estimated IMU bias at time $t = t_i$. Before exploring how this quantity can be updated given changes in estimated IMU bias, first consider it a fixed quantity. With $\Delta\hat{\mathcal{X}}_{ij}$ fixed, only $\Delta\mathcal{X}_{ij}(\mathcal{X}_i, \mathcal{X}_j)$ depends on the states at times t_i and t_j . The following section will derive how to compute the Jacobians of the residual with respect to $\mathcal{X}_i$ and $\mathcal{X}_j$, which are required to implement the preintegrated residual within batch estimation.

4.1 Jacobian Derivation

This section will showcase how to derive the Jacobians of the preintegrated residual shown in (135) with respect to the states $\mathcal{X}_i$ and $\mathcal{X}_j$, needed for batch estimation. The desired Jacobian can be found utilizing the chain rule, as

$$\frac{D\mathbf{e}_{ij}}{D\mathcal{X}_i} = \frac{D\mathbf{e}_{ij}}{D\Delta\mathcal{X}_{ij}} \frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_i}, \quad (138)$$

$$\frac{D\mathbf{e}_{ij}}{D\mathcal{X}_j} = \frac{D\mathbf{e}_{ij}}{D\Delta\mathcal{X}_{ij}} \frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_j}. \quad (139)$$

The first Jacobian is the Jacobian of the preintegrated residual with respect to the RMI, which can be computed as

$$\frac{D\mathbf{e}_{ij}}{D\Delta\mathcal{X}_{ij}} = \frac{D(\mathcal{Y} \ominus \mathcal{X})}{D\mathcal{Y}}. \quad (140)$$

For a **right** definition of the $\ominus$ operator, this Jacobian is given by

$$\frac{D(\mathcal{Y} \ominus \mathcal{X})}{D\mathcal{Y}} = \mathbf{J}_r^{-1}(\boldsymbol{\tau}), \quad (141)$$

where $\boldsymbol{\tau} = \mathcal{Y} \ominus \mathcal{X}$. Similarly, for a **left** definition of the $\ominus$ operator, this Jacobian is given by

$$\frac{D(\mathcal{Y} \ominus \mathcal{X})}{D\mathcal{X}} = \mathbf{J}_l^{-1}(\boldsymbol{\tau}). \quad (142)$$

To derive the Jacobians of the RMI $\Delta\mathcal{X}_{ij}$ with respect to the state $\mathcal{X}_i$ and $\mathcal{X}_j$, it is easiest to consider the exact expressions separately for a left and right perturbation.

4.1.1 Left Perturbation Residual Jacobians

This section will derive the Jacobians given by

$$\frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_i}, \quad \frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_j}, \quad (143)$$

that is, the Jacobian of the RMI $\Delta\mathcal{X}_{ij}$ with respect to the states at times $t = t_i$ and $t = t_j$. Note that $\Delta\mathcal{X}_{ij}$, $\mathcal{X}_i$, and $\mathcal{X}_j$ are all elements of $SE_2(3) \times \mathbb{R}^6$, and so care must be taken when computing these Jacobians, as this is a Jacobian of a function that maps from $SE_2(3) \times \mathbb{R}^6 \rightarrow SE_2(3) \times \mathbb{R}^6$. Consider first the function given by

$$\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) = (\Gamma_{ij} \Phi_{ij}(\mathbf{T}_i))^{-1} \mathbf{T}_j \quad (144)$$

First, consider the Jacobians of this function with respect to both $\mathbf{T}_i$ and $\mathbf{T}_j$. Perturbing $\mathbf{T}_i$ utilizing a left perturbation yields

$$\Upsilon_{ij}(\text{Exp}(\delta\xi_i^I) \bar{\mathbf{T}}_i, \bar{\mathbf{T}}_j) = (\Gamma_{ij} \Phi(\text{Exp}(\delta\xi_i^I) \bar{\mathbf{T}}_i))^{-1} \bar{\mathbf{T}}_j, \quad (145)$$

$$= (\Gamma_{ij} \text{Exp}(\mathbf{F}_{\Delta t} \delta\xi_i^I) \Phi(\bar{\mathbf{T}}_i))^{-1} \bar{\mathbf{T}}_j, \quad (146)$$

$$= \Phi(\bar{\mathbf{T}}_i)^{-1} \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \Gamma_{ij}^{-1} \bar{\mathbf{T}}_j, \quad (147)$$

$$\text{Exp}(\delta\xi^Y) \bar{\Upsilon}_{ij} = \Phi(\bar{\mathbf{T}}_i)^{-1} \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \Phi(\bar{\mathbf{T}}_i) \underbrace{\Phi(\bar{\mathbf{T}}_i)^{-1} \Gamma_{ij}^{-1} \bar{\mathbf{T}}_j}_{\bar{\Upsilon}_{ij}}, \quad (148)$$

$$\text{Exp}(\delta\xi^Y) = \Phi(\bar{\mathbf{T}}_i)^{-1} \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \Phi(\bar{\mathbf{T}}_i), \quad (149)$$

$$= \text{Exp}(-\text{Ad}(\Phi(\mathbf{T}_i)^{-1}) \mathbf{F}_{\Delta t} \delta\xi_i^I), \quad (150)$$

$$\delta\xi^Y = -\text{Ad}(\Phi(\mathbf{T}_i)^{-1}) \mathbf{F}_{\Delta t} \delta\xi_i^I, \quad (151)$$

where the key property of the adjoint $\mathcal{X} \text{Exp}(\boldsymbol{\tau}) \mathcal{X}^{-1} = \text{Exp}(\text{Ad}(\mathcal{X}) \boldsymbol{\tau})$ has been used. Hence, the Jacobian of Υ_{ij} with respect to $\mathbf{T}_i$ is given by

$$\frac{D\Upsilon_{ij}}{D\mathbf{T}_i} = -\text{Ad}(\Phi(\mathbf{T}_i)^{-1}) \mathbf{F}_{\Delta t}. \quad (152)$$

Similarly, the Jacobian of Υ_{ij} with respect to $\mathbf{T}_j$ can be found by perturbing both sides as

$$\text{Exp}(\delta\xi^Y) \tilde{\Upsilon}_{ij} = (\Gamma_{ij} \Phi(\mathbf{T}_i))^{-1} \text{Exp}(\delta\xi_j^I) \bar{\mathbf{T}}_j, \quad (153)$$

$$= (\Gamma_{ij} \Phi(\mathbf{T}_i))^{-1} \mathbf{T}_j \mathbf{T}_j^{-1} \text{Exp}(\delta\xi_j^I) \bar{\mathbf{T}}_j, \quad (154)$$

$$\text{Exp}(\delta\xi^Y) = (\Gamma_{ij} \Phi(\bar{\mathbf{T}}_i))^{-1} \text{Exp}(\delta\xi) (\Gamma_{ij} \Phi(\mathbf{T}_i)) \quad (155)$$

$$= \text{Exp}\left(\text{Ad}\left(\Phi(\bar{\mathbf{T}}_i)^{-1} \Gamma_{ij}^{-1}\right) \delta\xi\right), \quad (156)$$

$$\delta\xi^Y = \text{Ad}\left(\Phi(\bar{\mathbf{T}}_i)\right)^{-1} \Gamma_{ij}^{-1} \delta\xi_j^I, \quad (157)$$

meaning that the Jacobian of Υ_{ij} with respect to $\mathbf{T}_j$ is given by

$$\frac{D\Upsilon_{ij}}{D\mathbf{T}_j} = \text{Ad}\left(\Phi(\bar{\mathbf{T}}_i)\right)^{-1} \Gamma_{ij}^{-1}. \quad (158)$$

Finally, the expressions for the Jacobians of $\Delta\mathbf{b}_{ij}$ with respect to $\mathbf{b}_i$ and $\mathbf{b}_j$ are given by

$$\frac{D\Delta\mathbf{b}_{ij}}{D\mathbf{b}_i} = -\mathbf{1}_{6 \times 6}, \quad \frac{D\Delta\mathbf{b}_{ij}}{D\mathbf{b}_j} = \mathbf{1}_{6 \times 6}. \quad (159)$$

Combining these results, the full Jacobians of $\Delta\mathcal{X}_{ij}$ with respect to the states $\mathcal{X}_i$ and $\mathcal{X}_j$ are given by

$$\frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_i} = \begin{bmatrix} \mathbf{H}_{\mathcal{X}_i} & \mathbf{0}_{9 \times 6} \\ \mathbf{0}_{6 \times 9} & -\mathbf{1}_{6 \times 6} \end{bmatrix} \in \mathbb{R}^{15 \times 15} \quad (160)$$

$$\frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_j} = \begin{bmatrix} \mathbf{H}_{\mathcal{X}_j} & \mathbf{0}_{9 \times 6} \\ \mathbf{0}_{6 \times 9} & \mathbf{1}_{6 \times 6} \end{bmatrix} \in \mathbb{R}^{15 \times 15}, \quad (161)$$

with $\mathbf{H}_{\mathcal{X}_i}$ and $\mathbf{H}_{\mathcal{X}_j}$ defined in (152) and (158), respectively.

4.1.2 Right Perturbation Residual Jacobians

Next, the Jacobians of the RMI with respect to the states at times $t = t_i$ and $t = t_j$ will be derived for a right perturbation. Similarly to before, consider the function given by

$$\Upsilon_{ij}(\mathbf{T}_i, \mathbf{T}_j) = (\Gamma_{ij} \Phi_{ij}(\mathbf{T}_i))^{-1} \mathbf{T}_j \quad (162)$$

First, consider the Jacobians of this function with respect to $\mathbf{T}_i$ by perturbing both sides as

$$\tilde{\Upsilon}_{ij} \text{Exp}(\delta\xi^Y) = (\Gamma_{ij} \Phi_{ij}(\bar{\mathbf{T}}_i \text{Exp}(\delta\xi_i^I)))^{-1} \bar{\mathbf{T}}_j, \quad (163)$$

$$= \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \tilde{\Upsilon}_{ij}, \quad (164)$$

$$= \tilde{\Upsilon}_{ij} \tilde{\Upsilon}_{ij}^{-1} \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \tilde{\Upsilon}_{ij}, \quad (165)$$

$$\text{Exp}(\delta\xi^Y) = \tilde{\Upsilon}_{ij}^{-1} \text{Exp}(-\mathbf{F}_{\Delta t} \delta\xi_i^I) \tilde{\Upsilon}_{ij}, \quad (166)$$

$$\delta\xi^Y = -\text{Ad}(\tilde{\Upsilon}_{ij}^{-1}) \mathbf{F}_{\Delta t} \delta\xi_i^I, \quad (167)$$

meaning that the Jacobian of Υ_{ij} with respect to $\mathbf{T}_i$ is given by

$$\frac{D\Upsilon_{ij}}{D\mathbf{T}_i} = -\text{Ad}(\tilde{\Upsilon}_{ij}^{-1}) \mathbf{F}_{\Delta t}. \quad (168)$$

Next, the Jacobian of Υ_{ij} with respect to $\mathbf{T}_j$ is given by

$$\bar{\Upsilon}_{ij} \text{Exp}(\delta \boldsymbol{\xi}^Y) = (\boldsymbol{\Gamma}_{ij} \boldsymbol{\Phi}_{ij}(\bar{\mathbf{T}}_i))^{-1} \bar{\mathbf{T}}_j \text{Exp}(\delta \boldsymbol{\xi}_j^I), \quad (169)$$

$$\delta \boldsymbol{\xi}^Y = \delta \boldsymbol{\xi}_j^I, \quad (170)$$

meaning that the Jacobian of Υ_{ij} with respect to $\mathbf{T}_j$ is given by

$$\frac{D\Upsilon_{ij}}{D\mathbf{T}_j} = \mathbf{1}_{9 \times 9}. \quad (171)$$

The Jacobians of the bias RMI with respect to the bias states are the same as before, meaning that the full Jacobians utilizing a right perturbation can be written as

$$\frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_i} = \begin{bmatrix} \mathbf{H}_{\mathcal{X}_i} & \mathbf{0}_{9 \times 6} \\ \mathbf{0}_{6 \times 9} & -\mathbf{1}_{6 \times 6} \end{bmatrix} \in \mathbb{R}^{15 \times 15} \quad (172)$$

$$\frac{D\Delta\mathcal{X}_{ij}}{D\mathcal{X}_j} = \begin{bmatrix} \mathbf{H}_{\mathcal{X}_j} & \mathbf{0}_{9 \times 6} \\ \mathbf{0}_{6 \times 9} & \mathbf{1}_{6 \times 6} \end{bmatrix} \in \mathbb{R}^{15 \times 15}, \quad (173)$$

where $\mathbf{H}_{\mathcal{X}_i}$ and $\mathbf{H}_{\mathcal{X}_j}$ are defined in (168) and (171), respectively.

4.2 First-Order Bias Update

Note that the preintegrated RMI computed from measurements, $\Delta\mathcal{X}_{ij}^Y$, is recursively computed using the bias estimate during the mean propagation, denoted $\hat{\mathbf{b}}_i$. During optimization, the estimated value of the bias will change, as the bias is a state for estimation. Rather than repeat the entire preintegration each time the estimated bias is updated, the value of the RMI can be updated using a first-order approximation. The original preintegration papers [5] performs this operations on $SO(3) \times \mathbb{R}^6$, whereas the method shown here performs the bias update directly on $SE_2(3)$, similarly to [6, 9–11]. Consider the Jacobian given by

$$\frac{D\Delta\mathcal{X}_{ij}^Y}{D\mathcal{X}_i} \triangleq \mathbf{J}_{ij}^{\Delta\mathcal{X}} \in \mathbb{R}^{15 \times 15}, \quad (174)$$

which can be computed recursively starting from the initial condition $\mathbf{J}_{ii} = \mathbf{1} \in \mathbb{R}^{15 \times 15}$ as

$$\mathbf{J}_{ki+1}^{\Delta\mathcal{X}} = \mathbf{F}_k^{\Delta\mathcal{X}} \mathbf{J}_{ki}^{\Delta\mathcal{X}}, \quad (175)$$

where $\mathbf{F}_k^{\Delta\mathcal{X}}$ is the discrete-time Jacobian of the RMI with respect to the state computed during preintegration. With this Jacobian, given a new bias estimate $\hat{\mathbf{b}}_i$, the RMI can be updated as

$$\Delta\mathbf{X}_{ij}^{Y+} \approx \Delta\mathbf{X}_{ij}^Y(\bar{\mathbf{b}}_i) \oplus (\mathbf{J}_{ij}^b \delta\mathbf{b}), \quad (176)$$

where $\delta\mathbf{b} = \bar{\mathbf{b}}_i - \hat{\mathbf{b}}_i$ is the change in bias from the initial value used to compute $\Delta\mathbf{X}_{ij}^Y$, and $\mathbf{J}_{ij}^b \in \mathbb{R}^{9 \times 6}$ is the upper-right subblock of the full Jacobian $\mathbf{J}_{ij}^{\Delta\mathcal{X}}$. Note that the definition of the $\oplus$ operator should match the definitions of the $\oplus$ operator for the state and RMI. Utilizing the first order bias update, the full residual can be computed as

$$\mathbf{e}_{ij}(\mathcal{X}_i, \mathcal{X}_j) = \Delta\mathcal{X}_{ij}^X(\mathcal{X}_i, \mathcal{X}_j) \ominus \Delta\mathcal{X}_{ij}^{Y+}(\mathbf{b}_i), \quad (177)$$

where now, the updated measured RMI, $\Delta\mathcal{X}_{ij}^{Y+}$, is a function of the IMU bias. Thus, incorporating this residual in batch estimation requires computing extra additional Jacobians utilizing the chain rule.

References

- [1] P. Geneva, K. Eickenhoff, W. Lee, Y. Yang, and G. Huang, “Openvins: A Research Platform for Visual-Inertial Estimation,” in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2020, pp. 4666–4672.
- [2] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, “Fast-lio2: Fast Direct LiDAR-Inertial Odometry,” *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2053–2073, 2022.
- [3] J. Michalczyk, R. Jung, and S. Weiss, “Tightly-Coupled EKF-Based Radar-Inertial Odometry,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 12 336–12 343.
- [4] K. Li, Z. Cao, and U. D. Hanebeck, “Continuous-Time Ultra-Wideband-Inertial Fusion,” *IEEE Robotics and Automation Letters*, vol. 8, no. 7, pp. 4338–4345, 2023.
- [5] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, “On-Manifold Preintegration for Real-Time Visual-Inertial Odometry,” *IEEE Transactions on Robotics*, vol. 33, no. 1, pp. 1–21, 2016.
- [6] M. Brossard, A. Barrau, P. Chauchat, and S. Bonnabel, “Associating Uncertainty to Extended Poses for on Lie Group IMU Preintegration with Rotating Earth,” *IEEE Transactions on Robotics*, vol. 38, no. 2, pp. 998–1015, 2021.
- [7] J. Sola, J. Deray, and D. Atchuthan, “A Micro Lie Theory for State Estimation in Robotics,” *arXiv preprint arXiv:1812.01537*, 2018.
- [8] M. Hartley, “Contact-Aided State Estimation on Lie Groups for Legged Robot Mapping and Control,” PhD thesis, 2019.
- [9] S.-H. Tsao and S.-S. Jan, “Analytic IMU Preintegration That Associates Uncertainty on Matrix Lie Groups for Consistent Visual-Inertial Navigation Systems,” *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3820–3827, 2023.
- [10] G. Delama, A. Fornasier, R. Mahony, and S. Weiss, “Equivariant IMU Preintegration with Biases: A Galilean Group Approach,” *IEEE Robotics and Automation Letters*, 2024.
- [11] D. Lisus, M. Cohen, and J. R. Forbes, “Know What You Don’t Know: Consistency in Sliding Window Filtering with Unobservable States Applied to Visual-Inertial SLAM,” *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3382–3389, 2023.
- [12] J. A. Hesch, D. G. Kottas, S. L. Bowman, and S. I. Roumeliotis, “Consistency Analysis and Improvement of Vision-Aided Inertial Navigation,” *IEEE Transactions on Robotics*, vol. 30, no. 1, pp. 158–176, 2013.
- [13] “Local observability matrix and its application to observability analysis.”
- [14] “Consistent ekf-based visual-inertial odometry on matrix lie group,”

A Continuous-Time IMU Kinematics Jacobian Derivations

In this section, the Jacobians of the continuous-time IMU kinematics will be derived for both a left and right perturbation of the navigation state. Neglecting subscripts from brevity, recall that the continuous-time IMU kinematics are given by

$$\dot{\mathbf{C}} = \mathbf{C} (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g), \quad (178)$$

$$\dot{\mathbf{v}} = \mathbf{C} (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g}, \quad (179)$$

$$\dot{\mathbf{r}} = \mathbf{v}, \quad (180)$$

$$\dot{\mathbf{b}}^g = \mathbf{w}^b, \quad (181)$$

$$\dot{\mathbf{b}}^a = \mathbf{w}^a. \quad (182)$$

The next sections will derive the error dynamics of this continuous-time process model for both a left and right perturbation of the state.

A.1 Left Perturbation Derivation

To start, consider the state definition $\mathcal{X} = (\mathbf{T}, \mathbf{b}) \in SE_2(3) \times \mathbb{R}^6$, and the perturbation given by

$$\mathcal{X} = \bar{\mathcal{X}} \oplus \delta\boldsymbol{\xi} = (\bar{\mathbf{T}} \oplus \delta\boldsymbol{\xi}^{\text{nav}}, \bar{\mathbf{b}} + \delta\boldsymbol{\xi}^b). \quad (183)$$

Expanding the navigation state perturbation yields

$$\mathbf{T} = \text{Exp}(\delta\boldsymbol{\xi}^{\text{nav}}) \bar{\mathbf{T}}, \quad (184)$$

$$\begin{bmatrix} \mathbf{C} & \mathbf{v} & \mathbf{r} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} \delta\mathbf{C} & \delta\mathbf{v} & \delta\mathbf{r} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} \begin{bmatrix} \bar{\mathbf{C}} & \bar{\mathbf{v}} & \bar{\mathbf{r}} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (185)$$

$$= \begin{bmatrix} \delta\mathbf{C}\bar{\mathbf{C}} & \delta\mathbf{C}\bar{\mathbf{v}} + \delta\mathbf{v} & \delta\mathbf{C}\bar{\mathbf{r}} + \delta\mathbf{r} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}. \quad (186)$$

Rearranging yields the individual error definitions for the navigation state as

$$\delta\mathbf{C} = \mathbf{C}\bar{\mathbf{C}}^\top, \quad (187)$$

$$\delta\mathbf{v} = \mathbf{v} - \delta\mathbf{C}\bar{\mathbf{v}}, \quad (188)$$

$$\delta\mathbf{r} = \mathbf{r} - \delta\mathbf{C}\bar{\mathbf{r}}. \quad (189)$$

The error dynamics of the continuous-time IMU kinematics can now be derived by taking the time derivative of the error definitions. Starting with the attitude yields

$$\delta\dot{\mathbf{C}} = \dot{\mathbf{C}}\bar{\mathbf{C}}^\top + \mathbf{C}\dot{\bar{\mathbf{C}}}^\top \quad (190)$$

$$= \mathbf{C} (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times \bar{\mathbf{C}}^\top - \mathbf{C} (\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \bar{\mathbf{C}}^\top, \quad (191)$$

$$= \mathbf{C} (-\mathbf{b}^g + \bar{\mathbf{b}}^g - \mathbf{w}^g)^\times \bar{\mathbf{C}}^\top, \quad (192)$$

$$= \delta\mathbf{C}\bar{\mathbf{C}} (-\delta\mathbf{b}^g - \mathbf{w}^g)^\times \bar{\mathbf{C}}^\top, \quad (193)$$

$$= \delta\mathbf{C} (\bar{\mathbf{C}} (-\delta\mathbf{b}^g - \mathbf{w}^g))^\times \quad (194)$$

Linearizing by letting $\delta \mathbf{C} \approx \mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}$ and neglecting products of small terms yields

$$\delta \dot{\boldsymbol{\xi}}^{\phi^\times} \approx \left(\mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times} \right) \left(\bar{\mathbf{C}} \left(-\delta \boldsymbol{\xi}^{b_g} - \mathbf{w}^g \right) \right)^\times \quad (195)$$

$$\approx -\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{C}} \delta \mathbf{w}^g. \quad (196)$$

Next, consider the velocity error dynamics given by

$$\delta \dot{\mathbf{v}} = \dot{\mathbf{v}} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{v}} - \delta \mathbf{C} \dot{\bar{\mathbf{v}}} \quad (197)$$

$$= \mathbf{C} (\mathbf{u}^a - \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{v}} - \delta \mathbf{C} (\bar{\mathbf{C}} (\mathbf{u}^a - \bar{\mathbf{b}}^a) + \mathbf{g}) \quad (198)$$

$$= \delta \mathbf{C} \bar{\mathbf{C}} (\mathbf{u}^a - \bar{\mathbf{b}}^a - \delta \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{v}} - \delta \mathbf{C} (\bar{\mathbf{C}} (\mathbf{u}^a - \bar{\mathbf{b}}^a) + \mathbf{g}), \quad (199)$$

$$= \delta \mathbf{C} \bar{\mathbf{C}} (\delta \mathbf{b}^a - \mathbf{w}^a) + \mathbf{g} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{v}} - \delta \mathbf{C} \mathbf{g}_a. \quad (200)$$

Linearizing by letting $\delta \mathbf{C} \approx \mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}$, and $\delta \mathbf{v} \approx \delta \boldsymbol{\xi}^v$, and $\delta \dot{\bar{\mathbf{C}}} \approx (-\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{C}} \delta \mathbf{w}^g)^\times$ yields

$$\delta \dot{\boldsymbol{\xi}}^v \approx \left(\mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times} \right) \bar{\mathbf{C}} (-\delta \boldsymbol{\xi}^{b_a} - \delta \mathbf{w}^a) + \mathbf{g} - (-\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{C}} \delta \mathbf{w}^g)^\times \bar{\mathbf{v}} - \left(\mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times} \right) \mathbf{g}, \quad (201)$$

$$= -\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_a} - \bar{\mathbf{C}} \delta \mathbf{w}^a - \bar{\mathbf{v}}^\times \bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{v}}^\times \bar{\mathbf{C}} \delta \mathbf{w}^g + \mathbf{g}^\times \delta \boldsymbol{\xi}^{\phi^\times}. \quad (202)$$

Next, consider error dynamics $\delta \dot{\mathbf{r}}$ given by

$$\delta \dot{\mathbf{r}} = \dot{\mathbf{r}} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{r}} - \delta \mathbf{C} \dot{\bar{\mathbf{r}}}, \quad (203)$$

$$= \mathbf{v} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{r}} - \delta \mathbf{C} \bar{\mathbf{v}}, \quad (204)$$

$$= \delta \mathbf{C} \bar{\mathbf{v}} + \delta \mathbf{v} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{r}} - \delta \mathbf{C} \bar{\mathbf{v}} \quad (205)$$

$$= \delta \mathbf{v} - \delta \dot{\bar{\mathbf{C}}} \bar{\mathbf{r}} \quad (206)$$

Linearizing by letting $\delta \mathbf{C} \approx \mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}$, $\delta \mathbf{v} \approx \delta \boldsymbol{\xi}^v$, and $\delta \dot{\bar{\mathbf{C}}} \approx (-\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{C}} \delta \mathbf{w}^g)^\times$ yields

$$\delta \dot{\boldsymbol{\xi}}^r \approx \delta \boldsymbol{\xi}^v - (-\bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{C}} \delta \mathbf{w}^g)^\times \bar{\mathbf{r}}, \quad (207)$$

$$= \delta \boldsymbol{\xi}^v - \bar{\mathbf{r}}^\times \bar{\mathbf{C}} \delta \boldsymbol{\xi}^{b_g} - \bar{\mathbf{r}}^\times \bar{\mathbf{C}} \delta \mathbf{w}^g. \quad (208)$$

Finally, linearizing the bias error dynamics is given by

$$\delta \dot{\mathbf{b}} = \dot{\mathbf{b}} - \dot{\bar{\mathbf{b}}} = \mathbf{w}, \quad (209)$$

$$\delta \dot{\mathbf{b}} \approx \delta \mathbf{w}. \quad (210)$$

Hence, the continuous-time process model Jacobians for a left perturbation of the navigation state is given by

$$\mathbf{F} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\bar{\mathbf{C}} & \mathbf{0} \\ \mathbf{g}^\times & \mathbf{0} & \mathbf{0} & -\bar{\mathbf{v}}^\times \bar{\mathbf{C}} & -\bar{\mathbf{C}} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & -\bar{\mathbf{r}}^\times \bar{\mathbf{C}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (211)$$

$$\mathbf{L} = \begin{bmatrix} \bar{\mathbf{C}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \bar{\mathbf{v}}^\times \bar{\mathbf{C}} & \bar{\mathbf{C}} & \mathbf{0} & \mathbf{0} \\ \bar{\mathbf{r}}^\times \bar{\mathbf{C}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \quad (212)$$

A.2 Right Perturbation

Next, the same state definition $\mathcal{X} = (\mathbf{T}, \mathbf{b}) \in SE_2(3) \times \mathbb{R}^6$ will be considered, but the perturbation on the navigation state will be a right perturbation given by

$$\mathbf{T} = \bar{\mathbf{T}} \text{Exp}(\delta \boldsymbol{\xi}^{\text{nav}}), \quad (213)$$

$$\begin{bmatrix} \mathbf{C} & \mathbf{v} & \mathbf{r} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{C}} & \bar{\mathbf{v}} & \bar{\mathbf{r}} \end{bmatrix} \begin{bmatrix} \delta \mathbf{C} & \delta \mathbf{v} & \delta \mathbf{r} \end{bmatrix} \quad (214)$$

$$= \begin{bmatrix} \bar{\mathbf{C}} \delta \mathbf{C} & \bar{\mathbf{C}} \delta \mathbf{v} + \bar{\mathbf{v}} & \bar{\mathbf{C}} \delta \mathbf{r} + \bar{\mathbf{r}} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}. \quad (215)$$

This leads to the individual error definitions given by

$$\delta \mathbf{C} = \bar{\mathbf{C}}^\top \mathbf{C}, \quad (216)$$

$$\delta \mathbf{v} = \bar{\mathbf{C}}^\top (\mathbf{v} - \bar{\mathbf{v}}), \quad (217)$$

$$\delta \mathbf{r} = \bar{\mathbf{C}}^\top (\mathbf{r} - \bar{\mathbf{r}}). \quad (218)$$

Starting with the attitude error dynamics yields

$$\delta \dot{\mathbf{C}} = \dot{\bar{\mathbf{C}}}^\top \mathbf{C} + \bar{\mathbf{C}}^\top \dot{\mathbf{C}} \quad (219)$$

$$= -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \bar{\mathbf{C}}^\top \mathbf{C} + \bar{\mathbf{C}}^\top \mathbf{C} (\mathbf{u}^g - \mathbf{b}^g - \mathbf{w}^g)^\times, \quad (220)$$

$$= -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \delta \mathbf{C} + \delta \mathbf{C} (\mathbf{u}^g - \bar{\mathbf{b}}^g - \delta \mathbf{b}^g - \mathbf{w}^g)^\times. \quad (221)$$

Linearizing by letting $\delta \mathbf{C} \approx \mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}$ and $\delta \mathbf{b}^g \approx \delta \boldsymbol{\xi}^{b_g}$ yields

$$\delta \dot{\boldsymbol{\xi}}^{\phi^\times} \approx -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times (\mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}) + (\mathbf{1} + \delta \boldsymbol{\xi}^{\phi^\times}) (\mathbf{u}^g - \bar{\mathbf{b}}^g - \delta \boldsymbol{\xi}^{b_g} - \mathbf{w}^g)^\times, \quad (222)$$

$$\delta \dot{\boldsymbol{\xi}}^\phi \approx -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \delta \boldsymbol{\xi}^{\phi^\times} - \delta \boldsymbol{\xi}^{b_g} - \delta \mathbf{w}^g. \quad (223)$$

Next, consider the error dynamics $\delta \dot{\mathbf{v}}$ given by

$$\delta \dot{\mathbf{v}} = \dot{\bar{\mathbf{C}}}^\top (\mathbf{v} - \bar{\mathbf{v}}) + \bar{\mathbf{C}}^\top (\dot{\mathbf{v}} - \dot{\bar{\mathbf{v}}}). \quad (224)$$

Subbing in the appropriate perturbations and linearizing yields

$$\delta \boldsymbol{\xi}^v \approx -(\mathbf{u}^a - \bar{\mathbf{b}}^a)^\times \delta \boldsymbol{\xi}^\phi - (\mathbf{u}^g - \mathbf{b}^g)^\times \delta \boldsymbol{\xi}^v - \delta \boldsymbol{\xi}^{b_a} - \delta \mathbf{w}^a. \quad (225)$$

Next, consider the error dynamics $\delta \dot{\mathbf{r}}$ given by

$$\delta \dot{\mathbf{r}} = \dot{\bar{\mathbf{C}}}^\top (\mathbf{r} - \bar{\mathbf{r}}) + \bar{\mathbf{C}}^\top (\dot{\mathbf{r}} - \dot{\bar{\mathbf{r}}}), \quad (226)$$

$$= -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \delta \mathbf{r} + \delta \mathbf{v} \quad (227)$$

$$\approx -(\mathbf{u}^g - \bar{\mathbf{b}}^g)^\times \delta \boldsymbol{\xi}^r + \delta \boldsymbol{\xi}^v. \quad (228)$$

The bias error dynamics are identical to the previous derivation. All together, utilizing a right perturbation for the navigation state yields the continuous-time Jacobians given by

$$\mathbf{F}_c = \begin{bmatrix} -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} & \mathbf{0} \\ -(\mathbf{u}^a - \mathbf{b}^a)^\times & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} & -\mathbf{1} \\ \mathbf{0} & \mathbf{1} & -(\mathbf{u}^g - \mathbf{b}^g)^\times & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_c = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}. \quad (229)$$

A.3 Discrete-Time State Transition Matrix Derivation

This section will derive the discrete-time state transition matrix $\dot{\Phi}_{k,1}$, specifically for a left perturbation of the state. The discrete-time state transition matrix is a key component of performing observability analyses for aided-inertial navigation systems, as it is used to compute the “local observability matrix” (see [12] and [13] for example).

Recall that the state transition matrix $\Phi_{k,1}$ satisfies the following matrix differential equation [12]

$$\dot{\Phi}_{k,1} = \mathbf{F}_c(t) \Phi_{k,1}, \quad \Phi_{1,1} = \mathbf{1}, \quad (230)$$

where utilizing a left perturbation of the navigation state, $\mathbf{F}_c$ has the following structure.

$$\mathbf{F}_c(t) = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{C} & \mathbf{0} \\ \mathbf{g}^\times & \mathbf{0} & \mathbf{0} & -\mathbf{v}^\times \mathbf{C} & -\mathbf{C} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & -\mathbf{r}^\times \mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (231)$$

$$(232)$$

where $\mathbf{C}(t)$, $\mathbf{v}(t)$ and $\mathbf{r}(t)$ are nominal time-varying values for the attitude, velocity, and position. Examining the block elements of $\dot{\Phi}_{k,1}$ allows us to determine an analytical solution for the state transition matrix. Consider the multiplication given by

$$\dot{\Phi}_{k,1} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & -\mathbf{r}^\times \mathbf{C} & \mathbf{0} \\ \mathbf{g}^\times & \mathbf{0} & \mathbf{0} & -\mathbf{v}^\times \mathbf{C} & -\mathbf{C} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Phi_{k,1}^{11} & \Phi_{k,1}^{12} & \Phi_{k,1}^{13} & \Phi_{k,1}^{14} & \Phi_{k,1}^{15} \\ \Phi_{k,1}^{21} & \Phi_{k,1}^{22} & \Phi_{k,1}^{23} & \Phi_{k,1}^{24} & \Phi_{k,1}^{25} \\ \Phi_{k,1}^{31} & \Phi_{k,1}^{32} & \Phi_{k,1}^{33} & \Phi_{k,1}^{34} & \Phi_{k,1}^{35} \\ \Phi_{k,1}^{41} & \Phi_{k,1}^{42} & \Phi_{k,1}^{43} & \Phi_{k,1}^{44} & \Phi_{k,1}^{45} \\ \Phi_{k,1}^{51} & \Phi_{k,1}^{52} & \Phi_{k,1}^{53} & \Phi_{k,1}^{54} & \Phi_{k,1}^{55} \end{bmatrix} \quad (233)$$

Since the last two rows of the process model Jacobian are zero,

$$\dot{\Phi}_{k,1}^{i*} = \mathbf{0}, \quad i = 4, 5. \quad (234)$$

Utilizing the initial condition $\Phi_{k,1} = \mathbf{1}$, $\dot{\Phi}_{k,1}^{i,i} = \mathbf{1}$ if $i = 4, 5$ and $\mathbf{0}$ otherwise. Thus, the state transition matrix can now be written as

$$\dot{\Phi}_{k,1} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & -\mathbf{r}^\times \mathbf{C} & \mathbf{0} \\ \mathbf{g}^\times & \mathbf{0} & \mathbf{0} & -\mathbf{v}^\times \mathbf{C} & -\mathbf{C} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Phi_{k,1}^{11} & \Phi_{k,1}^{12} & \Phi_{k,1}^{13} & \Phi_{k,1}^{14} & \Phi_{k,1}^{15} \\ \Phi_{k,1}^{21} & \Phi_{k,1}^{22} & \Phi_{k,1}^{23} & \Phi_{k,1}^{24} & \Phi_{k,1}^{25} \\ \Phi_{k,1}^{31} & \Phi_{k,1}^{32} & \Phi_{k,1}^{33} & \Phi_{k,1}^{34} & \Phi_{k,1}^{35} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \quad (235)$$

Following a similar approach, expanding the first block row of the state transition matrix yields

$$\dot{\Phi}_{k,1}^{11} = \mathbf{0}, \quad (236)$$

$$\dot{\Phi}_{k,1}^{14} = -\mathbf{C}, \quad (237)$$

Utilizing the initial condition $\Phi_{k,1}^{11} = \mathbf{1}$, we have

$$\Phi_{k,1}^{11} = \mathbf{1}, \quad (238)$$

$$\Phi_{k,1}^{14} = -\int_{t_1}^{t_k} \mathbf{C}(\tau) d\tau. \quad (239)$$

Next, consider expanding the expressions for the second block row of the state transition matrix as

$$\dot{\Phi}_{k,1}^{21} = \mathbf{g}_a^\times, \quad (240)$$

$$\dot{\Phi}_{k,1}^{22} = \mathbf{0}, \quad (241)$$

$$\dot{\Phi}_{k,1}^{24} = \mathbf{g}_a^\times \Phi_{k,1}^{14} - \mathbf{v}^\times \mathbf{C}, \quad (242)$$

$$\dot{\Phi}_{k,1}^{25} = -\mathbf{C}. \quad (243)$$

Integrating these from their initial conditions yields

$$\Phi_{k,1}^{21} = \Delta t \mathbf{g}_a^\times, \quad (244)$$

$$\Phi_{k,1}^{22} = \mathbf{1}, \quad (245)$$

$$\Phi_{k,1}^{24} = -\int_{t_1}^{t_k} \left(\mathbf{v}(\tau)^\times \mathbf{C}(\tau) - \mathbf{g}_a^\times \int_{t_1}^{\tau} \mathbf{C}(s) ds \right) d\tau, \quad (246)$$

$$\Phi_{k,1}^{25} = -\int_{t_1}^{t_k} \mathbf{C}(\tau) d\tau, \quad (247)$$

where $\Delta t = t_k - t_1$. Next, expanding the expressions for the third block row of the state transition matrix yields

$$\dot{\Phi}_{k,1}^{31} = \Phi_{k,1}^{21}, \quad (248)$$

$$\dot{\Phi}_{k,1}^{32} = \Phi_{k,1}^{22}, \quad (249)$$

$$\dot{\Phi}_{k,1}^{33} = \mathbf{0}, \quad (250)$$

$$\dot{\Phi}_{k,1}^{34} = \Phi_{k,1}^{24} - \mathbf{r}^\times \mathbf{C}, \quad (251)$$

$$\dot{\Phi}_{k,1}^{35} = \Phi_{k,1}^{25}. \quad (252)$$

Integrating these from their initial conditions yields

$$\Phi_{k,1}^{31} = \frac{1}{2} \mathbf{g}_a^\times \Delta t, \quad (253)$$

$$\Phi_{k,1}^{32} = \Delta t \mathbf{1}, \quad (254)$$

$$\Phi_{k,1}^{33} = \mathbf{1}, \quad (255)$$

$$\Phi_{k,1}^{34} = - \int_{t_1}^{t_k} (\mathbf{r}(\theta)^\times \mathbf{C}(\theta)) \quad (256)$$

$$+ \left[\int_{t_1}^{\theta} \left(\mathbf{v}(\tau)^\times \mathbf{C}(\tau) + \mathbf{g}_a^\times \int_{t_1}^{\tau} \mathbf{C}(s) ds \right) d\tau \right] d\theta, \quad (257)$$

$$\Phi_{k,1}^{35} = - \int_{t_1}^{t_k} \left(\int_{t_1}^{\tau} (\mathbf{C}(s) ds) \right) d\tau, \quad (258)$$

$$(259)$$

In summary, putting all of these components together, the discrete-time state transition matrix has the following structure.

$$\Phi_{k,1} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \Phi_{k,1}^{14} & \mathbf{0} & \mathbf{0} \\ \mathbf{g}_a^\times \Delta t & \mathbf{1} & \mathbf{0} & \Phi_{k,1}^{24} & \Phi_{k,1}^{25} & \mathbf{0} \\ \frac{1}{2} \mathbf{g}_a^\times \Delta t^2 & \Delta t \mathbf{1} & \mathbf{1} & \Phi_{k,1}^{34} & \Phi_{k,1}^{35} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}, \quad (260)$$

with the following form of the blocks

$$\Phi_{k,1}^{14} = - \int_{t_1}^{t_k} \mathbf{C}(\tau) d\tau \quad (261)$$

$$\Phi_{k,1}^{34} = - \int_{t_1}^{t_k} (\mathbf{r}(\theta)^\times \mathbf{C}(\theta)) \quad (262)$$

$$+ \left[\int_{t_1}^{\theta} \left(\mathbf{v}(\tau)^\times \mathbf{C}(\tau) + \mathbf{g}_a^\times \int_{t_1}^{\tau} \mathbf{C}(s) ds \right) d\tau \right] d\theta, \quad (263)$$

$$\Phi_{k,1}^{35} = - \int_{t_1}^{t_k} \left(\int_{t_1}^{\tau} (\mathbf{C}(s) ds) \right) d\tau, \quad (264)$$

$$\Phi_{k,1}^{24} = - \int_{t_1}^{t_k} \left(\mathbf{v}(\tau)^\times \mathbf{C}(\tau) + \mathbf{g}_a^\times \int_{t_1}^{\tau} \mathbf{C}(s) ds \right) d\tau, \quad (265)$$

$$\Phi_{k,1}^{25} = \Phi_{14} = - \int_{t_1}^{t_k} \mathbf{C}(\tau) d\tau. \quad (266)$$

This is consistent with the expressions found in [14].